

Project title : eWaste Recycling (B2C)

Description

Most people have eWaste - electronic waste - sitting in drawers at home, including old smartphones, laptops, tablets, consoles, televisions, etc.

There are many reasons these devices should be recycled as soon as possible, including:

- value is higher the sooner they are recycled, e.g. if sold
- more likely to be used, and also used for longer, by their next owners
- if broken, the parts they contain are more useful the sooner they are made available

There are also many reasons why people 'hang on' to devices, including

- concern that their data might be stolen
- seen as a backup if they lose their current device
- data is still on them, e.g. photos
- may be collectable

eWaste recycling facilities are available and also data cleansing facilities, but these often export the problem to other countries

This project proposes that owners can be offered a service that will retrieve data from devices as well as wiping them clean before they are then

- reused locally
- or broken into spare parts for use nationally or recycled ethically within the UK, or local to where the recycled materials will be reused

The project must offer a web hub where devices can be identified according to age and demand with an option for owners to pay for data retrieval. Where devices might be resold, e.g. newish smart phones, etc., then the hub should identify third parties where the owner might sell their device, with a referrer fee (potentially) paid by the third party for successful reselling. The seller would still be offered a paid option of having data retrieved and their device wiped.

The Hub will be a web application with web server/database with desktop usage being essential; a mobile browser interface is an option.

Out of scope

Selling devices, offering an eWaste service to Businesses.

Application users

Expected to include device owners, eWaste (company) staff

Expected Features

1. Standard account registration and login, optionally common third party accounts may be used (e.g. Google) but is not expected and is only recommended if you already have experience.
2. Accept details of devices and identify whether they are 'current' (currently popular), 'recycle', 'rare' or 'unknown'.
3. Current devices will have listings for expected value according to third parties, such as CeX, etc. with a focus on local places to trade in by physically handing in. Owner to be given a QRCode that offers a bonus (e.g. voucher) which also identifies the eWaste Hub for

a (potential) referral fee and simplifies them handing in a device (since they will have entered data already). Data wiping is guaranteed by the third party.

4. 'Recycle' devices are offered for recycling with data retrieval (at a fee) and data wiping (at no cost). Data retrieval devices are accepted by the eWaste Hub themselves for retrieving data which is then hosted in the cloud and a secure link emailed to the owner for a limited time (valid for 3 months, can be retrieved from 3-6 months with a further fee payment, then deleted). Data wiping will be handled by a third party who recycle the device.
5. Rare devices are handled in a similar manner to current devices but the third parties may include eBay, etc.
6. Unknown devices are flagged for employees to update the device database and contact the owner.
7. Expected roles will include (device) owner, admin and staff (employee).
8. For staff, creation of (new) device details. This should be initially populated from an offered device (by an owner). Draft details should be saved (but not visible).
9. Devices can be made visible or hidden, e.g. in case of incomplete/incorrect details.
10. Devices can be moved between classifications, which include current/recycle/rare/unwanted as well as device types, i.e. console/phone/tablet/laptop, etc. At least two device types should be shown working.
11. When a device is current, details for at least one third party should be shown, e.g. in a new/pop out window/tab.
12. Payments for data transfer should be via paypal and stripe (sandbox only).
13. An Admin dashboard will need to be available for managing users accounts including upgrading accounts for staff.
14. You should include relevant reports for staff, including payments, device processing (including status), referral fees.